



# Chapter 2

## Basic Structures: Sets, Functions, Sequences & Sums

MAD101

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# Sets

## 1 Sets

### Definition 1.1

A set is an unordered collection of distinct objects, called **elements or members** of the set. A set is said to contain its elements. We write  $a \in A$  to denote that  $a$  is an element of the set  $A$ . The notation  $a \notin A$  denotes that  $a$  is not an element of the set  $A$ .

### Example 1.1.

1.  $\{1, 2, \{3\}\}$  is the set containing “1” and “2” and “ $\{3\}$ ”.
2.  $\{1, 1, 2, 3, 3\} = \{1, 2, 3\} = \{3, 2, 1\}$ .
3.  $\emptyset = \{\}$  is the empty set, or the set containing no element.
4.  $\emptyset \neq \{\emptyset\}$ .

# Venn Diagrams

## 1 Sets

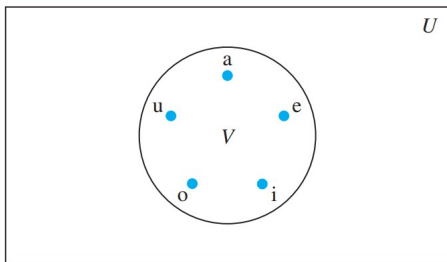
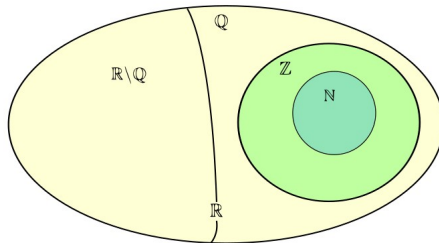


Figure: Venn diagram for the set of vowels.



1.  $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ , the set of all natural numbers.
2.  $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ , the set of all integers.
3.  $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$ , the set of all positive integers.
4.  $\mathbb{Q} = \{p/q | p \in \mathbb{Z}, q \in \mathbb{Z}\}$ , the set of all rational numbers.
5.  $\mathbb{R}$  is the set of real numbers.
6.  $\mathbb{C}$  is the set of all complex numbers.

# Subset ( $\subseteq$ ) & Proper subset ( $\subset$ )

1 Sets

$$A \subseteq B$$

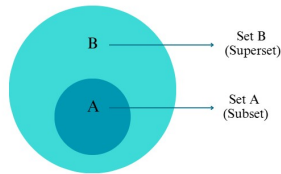
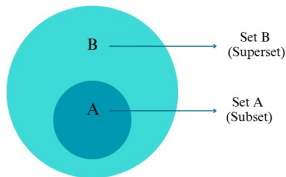
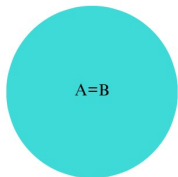
“A is a subset of B.”

$$\forall x((x \in A) \rightarrow (x \in B))$$

$$A \subset B$$

“A is a proper subset of B.”

$$A \subseteq B \text{ and } A \neq B$$



## Definition 1.2

Two sets are **equal** if and only if they have the same elements. Therefore, if  $A$  and  $B$  are sets, then  $A$  and  $B$  are equal if and only if  $\forall x((x \in A) \leftrightarrow (x \in B))$ . We write  $A = B$  if  $A$  and  $B$  are equal sets.

# Properties

## 1 Sets

**Note 1.1.** For every set  $S$ , (i )  $\emptyset \subseteq S$  and (ii )  $S \subseteq S$ .

**Example 1.2.** Determine whether each of these statements is true or false.

1.  $\{1, 2, 3\} \subseteq \{1, 2, 3, 4, 5\}$

2.  $\{1, 2, 3\} \subset \{1, 2, 3, 4, 5\}$

3.  $\emptyset \subseteq \{1, 2, 3\}$

4.  $\emptyset \in \{1, 2, 3\}$

5.  $\emptyset \subseteq \{\emptyset, 1, 2, 3\}$

6.  $\emptyset \in \{\emptyset, 1, 2, 3\}$

7.  $\{x\} \subseteq \{x\}$

8.  $\{x\} \in \{x, \{x\}\}$

9.  $\{x\} \subseteq \{x, \{x\}\}$

10.  $\{x\} \in \{x\}$



# Cardinality

## 1 Sets

### Definition 1.3

If  $S$  is finite, then the **cardinality** of  $S$ ,  $|S|$ , is the number of distinct elements in  $S$ .

### Definition 1.4

A set is said to be infinite if it is not finite.

### Example 1.3.

1.  $S = \{1, 2, 3\} \rightarrow |S|=3$
2.  $S = \{2, 4, 1, 7\} \rightarrow |S|=4$
3.  $S = \emptyset \rightarrow |S|=0$
4.  $S = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\} \rightarrow |S|=3$
5.  $S = \{1, 2, 3, \dots\} \rightarrow |S|$  is infinite

**Note 1.2.** When a infinite set  $S$  is countable, we denote the cardinality of  $S$  is  $|S| = \aleph_0$  (aleph null).

If  $S$  is uncountable and infinite, we denote the cardinality of  $S$  is  $|S| = 2^{\aleph_0}$ .



# Power sets

## 1 Sets

### Definition 1.5

Given a set  $S$ , the power set of  $S$  is the set of all subsets of the set  $S$ . The power set of  $S$  is denoted by  $P(S) \equiv 2^S$ .

**Note 1.3.** If  $S$  finite, then  $|2^S| = 2^{|S|}$  (If  $|S| = n$ , then  $|2^S| = 2^n$ )

### Example 1.4.

- If  $S = \{a\}$  then  $2^S = \{\emptyset, \{a\}\}$
- If  $S = \{a, b\}$  then  $2^S = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
- If  $S = \emptyset$  then  $2^S = \{\emptyset\}$
- If  $S = \{\emptyset, \{\emptyset\}\}$  then  $2^S = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$

**Example 1.5.** Given  $A = \{\emptyset, 1, 2, 3\}$ . Find the cardinality of  $P(A)$ .





# Cartesian Product

## 1 Sets

### Definition 1.6

The **Cartesian Product** of two sets  $A$  and  $B$  is  $A \times B = \{(a, b) : a \in A \wedge b \in B\}$ .

The **Cartesian Product** of  $n$  sets  $A_1, A_2, \dots, A_n$  is  $A_1 \times \dots \times A_n = \{(a_1, \dots, a_n) : a_i \in A_i, \forall i = \overline{1, n}\}$

### Note 1.4.

1.  $A \times B \neq B \times A$ .
2. If  $A, B$  are finite,  $|A \times B| = |A||B|$ .
3.  $A^n = A \times A \times A \times \dots \times A$ .

**Example 1.6.** Given  $A = \{a, b\}$ ,  $B = \{1, 2, 3\}$ ,  $C = \{0, 1\}$ . Find  $A \times B$ ,  $B \times A$ ,  $A \times B \times C$ ,  $A^2$ .

**Example 1.7.** Given  $A = \{0, \emptyset\}$ . Find the cardinality of  $P(A^2)$ .



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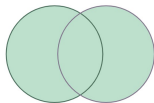
## 2 Set operations

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# Set operations

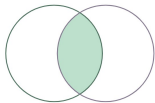
## 2 Set operations

Union



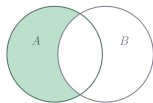
$$A \cup B$$

Intersection



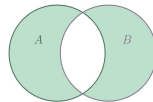
$$A \cap B$$

Difference



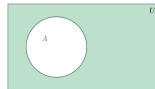
$$A - B$$

Symmetric  
Difference



$$A \oplus B$$

Complement



$$\bar{A}/A^c$$

### Definition 2.1

Two sets are called disjoint if their intersection is the empty set.



# Computer representation of sets

## 2 Set operations

The universal set  $U = \{x_1, x_2, \dots, x_n\}$ . Let  $A \subseteq U$ . Then the **bit string representation** of  $A$  is the bit string of length  $n$ :  $a_1 a_2 \dots a_n$  such that

$$a_i = \begin{cases} 1, & x_i \in A \\ 0, & \text{otherwise.} \end{cases}$$

**Example 2.1.** Let  $U = \{x_1, x_2, x_3, x_4, x_5, x_6\}$  and  $A = \{x_1, x_3, x_5, x_6\}$ .

$\hookrightarrow$  The bit string representation of  $A$  is 101011.

**Example 2.2.** Let  $U = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ . Given the subsets

$A = \{1, 2, 3, 4, 8\}$ ,  $B = \{0, 5, 6, 7, 9\}$ .

Find the bit string representing the subset  $A - B$ ,  $A \cap B$ ,  $A \cup B$ ,  $A \oplus B$ .



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# Functions

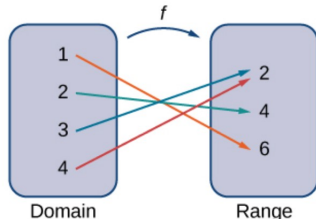
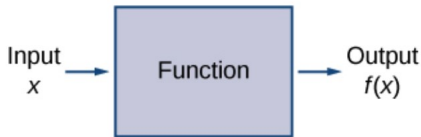
## 3 Functions

A function  $f$  consists of a set of inputs, a set of outputs, and a rule for assigning **each input to exactly one output**. We denote

$$f : A \longrightarrow B$$

$$x \longmapsto \underbrace{y = f(x)}_{\text{image}}$$

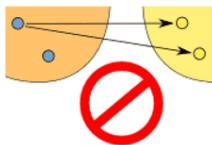
The **set of inputs** is called the **domain**  $A$  of the function. The **set of outputs** is called the **co-domain**  $B$  of the function.



A function can be visualized as an input/output device.

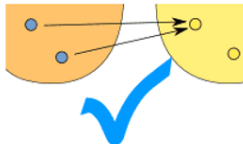
## Example

### 3 Functions



(one-to-many)

This is **NOT** OK in a function



(many-to-one)

But this **is** OK in a function

What are functions?

$$1. f : \mathbb{Z} \rightarrow \mathbb{R} : f(x) = x^2 + 2.$$

$$2. f : \mathbb{Z} \rightarrow \mathbb{R} : f(x) = \frac{1}{(x-1)^2} + 5x.$$

$$3. f : \mathbb{R} \rightarrow \mathbb{R} : f(x) = \frac{2x+5}{7}.$$

$$3. f : \mathbb{R} \rightarrow \mathbb{R} : f(x) = \frac{1}{x^2 - 2}.$$

$$4. f : \mathbb{Z} \rightarrow \mathbb{R} : f(x) = \frac{1}{x^2 - 2}.$$

$$5. f : \mathbb{Z} \rightarrow \mathbb{Z} : f(x) = \frac{1}{x^2 - 2}.$$



# Some Important Functions

## 3 Functions

**Ceiling.**  $f(x) = \lceil x \rceil$  the least integer  $y$  so that  $x \leq y$ .

### Example 3.1.

- a.  $\lceil 1.2 \rceil$
- b.  $\lceil -1.2 \rceil$
- c.  $\lceil 1 \rceil$

**Floor.**  $f(x) = \lfloor x \rfloor$  the greatest integer  $y$  so that  $y \leq x$ .

### Example 3.2.

- a.  $\lfloor 1.8 \rfloor$
- b.  $\lfloor -1.8 \rfloor$
- c.  $\lfloor -5 \rfloor$

Example 3.3.  $\lceil -1.1 + \lfloor 1.1 \rfloor \rceil = \dots$

Note 3.1.  $\boxed{\lfloor x \rfloor \leq x \leq \lceil x \rceil}.$





## Quiz

### 3 Functions

**Q1.** Let  $f$  be floor function and  $g$  be a ceiling function. Which of the following is true?

Select one:

a.  $f(-3.1)=-3$

b.  $g(-4.5)=-4$

c.  $g(7)=8$

d.  $f(5.3)=6$

**Q2.** Study relations in the set of real numbers  $\mathbb{R}$ .

(i)  $f : \mathbb{R} \rightarrow \mathbb{R} : f(x) = \frac{x+1}{x^2+3}$ .

(ii)  $f : \mathbb{R} \rightarrow \mathbb{R} : f(x) = \frac{x}{2x^2-6x-1}$ .

Select one:

a. (i) is not a function, (ii) is not a function.

b. (i) is a function, (ii) is a function.

c. (i) is a function, (ii) is not a function.

d. (i) is not a function, (ii) is a function.

# One-to-One Function/Injection

## 3 Functions

### Definition 3.1

A function  $f : A \rightarrow B$  is **one-to-one** if and only if

$$\forall x, y \in A, (x \neq y) \implies f(x) \neq f(y) \equiv \forall x, y \in A, f(x) = f(y) \implies x = y.$$

A function  $f : A \rightarrow B$  is **NOT one-to-one** if  $\exists x, y \in A, (x \neq y) \wedge (f(x) = f(y))$ .

**Example 3.4.** The following functions are one to one or not:

a.  $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^2$

b.  $f : [0, +\infty) \rightarrow \mathbb{R}, f(x) = x^2 + 1$

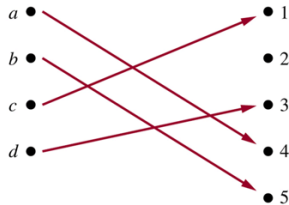


Figure: A one-to-one function.

# Onto Function/Surjection

## 3 Functions

### Definition 3.2

A function  $f : A \rightarrow B$  is **onto** if and only if  $\forall y \in B, \exists x \in A | f(x) = y$ .

**Example 3.5.** The following functions are onto or not

- $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x^2$
- $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = x + 1$

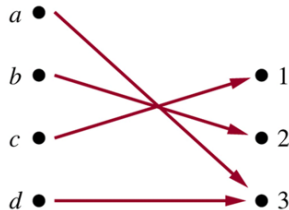


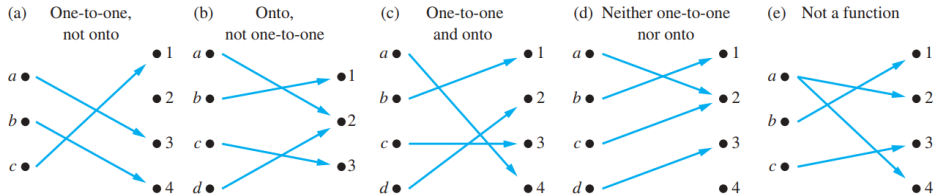
Figure: An onto function.

# Bijection/One-to-one correspondence

## 3 Functions

### Definition 3.3

The function  $f$  is a **one-to-one correspondence**, or a **bijection**, if it is **both one-to-one and onto**.



**Example 3.6.** The following functions are bijection or not

1.  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x + 1$
2.  $f : \mathbb{Z} \rightarrow \mathbb{N}, f(x) = (2 - x)^2$

# Inverse Functions

## 3 Functions

### Definition 3.4

Let  $f : A \rightarrow B$  be a **bijection**. Then the **inverse function** of  $f$ , denoted by  $f^{-1}$  is the function that assigns each element  $y$  in  $B$  the unique element  $x$  in  $A$  such that  $f(x) = y$ . Thus  $f^{-1}(y) = x$ .

$$f^{-1} : B \longrightarrow A$$

$$y \longmapsto f^{-1}(y) = x$$

**Example 3.7.** Is the function

$f(x) = x + 1$  from  $\mathbb{R}$  to  $\mathbb{R}$  invertible?

What is its inverse?

# Inverse Functions

## 3 Functions

### Definition 3.5

Let  $f : A \rightarrow B$  be a **bijection**. Then the **inverse function** of  $f$ , denoted by  $f^{-1}$  is the function that assigns each element  $y$  in  $B$  the unique element  $x$  in  $A$  such that  $f(x) = y$ . Thus  $f^{-1}(y) = x$ .

$$f^{-1} : B \rightarrow A$$

$$y \mapsto f^{-1}(y) = x$$

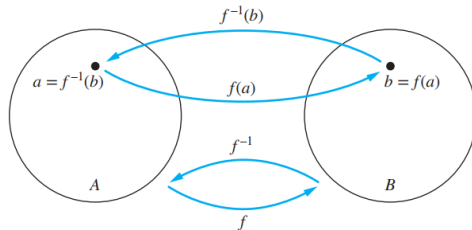
**Example 3.8.** Is the function  $f(x) = x + 1$  from  $\mathbb{R}$  to  $\mathbb{R}$  invertible? What is its inverse?

### Solution.

$f : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = x + 1$  is bijection.

$\implies \exists f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f^{-1}(y) = y - 1$ .

We also write  $f^{-1}(x) = x - 1$ .

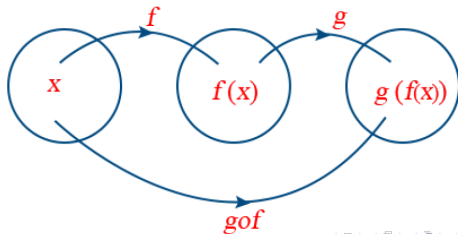


# Function composition

## 3 Functions

### Definition 3.6

Consider the function  $f : A \rightarrow B$ , and the function  $g : D \rightarrow E$  with domain  $D$  and range  $E$ . If  $B \subset D$ , then the composite function  $(g \circ f)(x)$  is the function with domain  $A$  such that  $(g \circ f)(x) = g(f(x))$ .



### Example 3.9. Given

$$f(x) = x^2 + 1, g(x) = 1/x.$$

1. Find  $(g \circ f)(x)$ ,  $(f \circ g)(x)$  and state its domain and range.
2. Evaluate  $(g \circ f)(4)$ ,  $(g \circ f)(-1/2)$ .

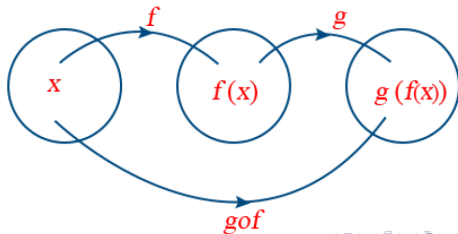
**Note 3.2.**  $f \circ g \neq g \circ f$ .

# Function composition

## 3 Functions

### Definition 3.7

Consider the function  $f : A \rightarrow B$ , and the function  $g : D \rightarrow E$  with domain  $D$  and range  $E$ . If  $B \subset D$ , then the composite function  $(g \circ f)(x)$  is the function with domain  $A$  such that  $(g \circ f)(x) = g(f(x))$ .



### Example 3.10. Given

$$f(x) = x^2 + 1, g(x) = 1/x.$$

1. Find  $(g \circ f)(x)$ ,  $(f \circ g)(x)$  and state its domain and range.
2. Evaluate  $(g \circ f)(4)$ ,  $(g \circ f)(-1/2)$ .

### Solution.

$$(g \circ f)(x): D = \mathbb{R}, \text{ the range is } (0, 1].$$

$$(f \circ g)(x): D = \mathbb{R} \setminus \{0\}, \text{ the range is } (1, +\infty)$$

**Note 3.3.**  $f \circ g \neq g \circ f$ .





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# Sequences

## 4 Sequences

### Definition 4.1

A **sequence** is a function from a subset of the set of integers (usually either the set  $\{0, 1, 2, \dots\}$  or the set  $\{1, 2, 3, \dots\}$ ) to a set  $S$ . We use the notation  $a_n$  to denote the image of the integer  $n$ . We call  $a_n$  a term of the sequence.

$$f : \mathbb{N} \longrightarrow \mathbb{R}$$

$$n \longmapsto f(n) = a_n$$

The list of the terms of this sequence, beginning with  $a_1$ , namely,  $a_1, a_2, a_3, a_4, \dots, a_{n-1}, a_n, a_{n+1}, \dots$

### Example 4.1.

1. Consider the sequence  $\{a_n\}$ , where  $a_n = \frac{1}{n} \implies 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$
2. **Geometric progression:**  $a, ar, ar^2, \dots, ar^n, \dots$  (**r: common ratio**).
3. **Arithmetic progression:**  $a, a + d, a + 2d, \dots, a + nd, \dots$  (**d: common difference**).
4. **Constant progression:**  $c, c, c, c, c, \dots$  ( $c \in \mathbb{R}$ ).



# Recurrence Relations

## 4 Sequences

### Definition 4.2

A **recurrence relation** for the sequence  $\{a_n\}$  is an equation that expresses  $a_n$  in terms of one or more of the previous terms of the sequence, namely,  $a_0, a_1, \dots, a_{n-1}, \forall n \in \mathbb{Z} (n \geq n_0, n_0 \in \mathbb{Z}^+)$ . A **sequence** is called a **solution** of a recurrence relation if its terms satisfy the recurrence relation. (A recurrence relation is said to recursively define a sequence. We will explain this alternative terminology in Chapter 5.)

**Example 4.2.** Let  $\{a_n\}$  be a sequence that satisfies the recurrence relation  $a_n = a_{n-1} + 3$  for  $n = 1, 2, 3, \dots$ , and suppose that  $a_0 = 2$ . What are  $a_1, a_2$ , and  $a_3$ ?

### Definition 4.3

The **Fibonacci sequence**,  $f_0, f_1, f_2, \dots$ , is defined by the initial conditions  $f_0 = 0, f_1 = 1$ , and the recurrence relation  $f_n = f_{n-1} + f_{n-2}$  for  $n = 2, 3, 4, \dots$

**Example 4.3.** Find the Fibonacci numbers  $f_2, f_3, f_4, f_5$ , and  $f_6$ .



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# Summations

## 5 Summations

- **Notation.**  $\sum_{i=1}^k a_i = a_1 + a_2 + \dots + a_k$
- **Properties.**
  1.  $\sum_{i=1}^k (ca_i + db_i) = c \sum_{i=1}^k a_i + d \sum_{i=1}^k b_i$
  2.  $\sum_{i=1}^k a = a + a + \dots + a = ka$

### Note 5.1.

- $\sum_{k=1}^n k = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$
- $\sum_{k=0}^n ar^k = a \frac{1 - r^{n+1}}{1 - r} (r \neq 1)$

**Example 5.1.** What is the value of

a.  $\sum_{i=1}^4 (i+2)^2$

b.  $\sum_{i=1}^6 2$

c.  $\sum_{k=1}^{50} 2^k$

d.  $\prod_{i=1}^{1000} (-1)^i$

e.  $\sum_{i=1}^3 \sum_{j=2}^4 ij$



Q&A

*Thank you for listening!*